

QUESTION-13

Constructing a Binary Tree

main.c	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 struct Node { 4 int data; 5 struct Node *next; 6 }; 7 struct Node *front = NULL; 8 struct Node *rear = NULL; 9 void enqueue(int value) { 10 struct Node *newNode; 11 newNode = (struct Node *)malloc(sizeof 12 (struct Node)); 13 newNode->data = value; 14 newNode->next = NULL; 15 if (rear == NULL) { 16 front = rear = newNode; 17 } else { 18 rear->next = newNode; 19 rear = newNode; 20 } 21 printf("%d inserted into queue\n", value); 22 } 23 void dequeue() { 24 struct Node *temp;</pre>	<pre>10 inserted into queue 20 inserted into queue 30 inserted into queue Queue Elements: 10 20 30 Front Element: 10 Deleted Element: 10 Queue Elements: 20 30 === Code Execution Successful ===</pre>

main.c	Output
<pre>24 if (front == NULL) { 25 printf("Queue Underflow\n"); 26 return; 27 } 28 temp = front; 29 printf("Deleted Element: %d\n", front 30 ->data); 31 front = front->next; 32 if (front == NULL) 33 rear = NULL; 34 free(temp); 35 } 36 void peek() { 37 if (front == NULL) 38 printf("Queue is Empty\n"); 39 else 40 printf("Front Element: %d\n", front 41 ->data); 42 } 43 void display() { 44 struct Node *temp; 45 if (front == NULL) { 46 printf("Queue is Empty\n"); 47 return; 48 }</pre>	<pre>10 inserted into queue 20 inserted into queue 30 inserted into queue Queue Elements: 10 20 30 Front Element: 10 Deleted Element: 10 Queue Elements: 20 30 === Code Execution Successful ===</pre>